Inference problems

1. $SE = \frac{10}{\sqrt{100}} = 1.0 \quad z = \frac{52-50}{1} = 2$

$\alpha = .10: 2.0 > 1.645$ → reject

$\alpha = .05: 2.0 > 1.96$ → reject

$\alpha = .01: 2.0 < 2.576$ → fail to reject

$52 \pm 1.96(1) = (50.04, 53.96)$

2. $z = \frac{3.1}{1.3} = 2.38$

90%: (0.96, 5.24) → reject

95%: (0.55, 5.65) → reject

99%: (-0.25, 6.45) → fail to reject

rejects @ $\alpha = .10$ & $.05$ and fail to reject @ $\alpha = .01$

3. $SE = \frac{3}{\sqrt{36}} = .5 \quad z = \frac{11.5-12}{0.5} = -1$

95% CI

$11.5 \pm 1.96(0.5) = (10.52, 12.48)$

$|-1.0| < 1.96$ but has 12 so we fail to reject & 12 is a plausible value

4. $SE = \sqrt{\frac{64}{100} + \frac{49}{100}} = 1.063$

$z = 2.82$

$3 \pm 1.96(1.063) = (0.92, 5.08)$

no zero so means differ

5. $SE = \sqrt{\frac{36}{81} + \frac{25}{64}} = 0.914$ diff = 2

$z = 2.19$

90%	.10	1.645	$2.189 > 1.645$ reject
95%	.05	1.96	$2.189 > 1.96$ reject
99%	.01	2.576	$2.189 < 2.576$ fail to reject

$2 \pm 1.96(0.9138) = (0.21, 3.79)$

6. $SE = \sqrt{\frac{2.25}{49} + \frac{1.96}{64}} = 0.2767$

$z = \frac{0.4-0}{0.2767} = 1.446 < 1.96$ fail to reject H_0

$0.4 \pm 1.96(0.267) = (-0.142, 0.942)$

the differences aren't large when looking at the noise but there still isn't enough info to distinguish anything at the 5% level